

Next Generation Data Centers and Energy Management Systems

Lecture 4B: Applied Computational Fluid Dynamics (CFD)

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Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- **Applied Computational Fluid Dynamics (CFD)**
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- Machine Learning and Optimization Tools
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Derive the “temperature formulation” of the first law of thermodynamics (conservation of energy).
- Use a finite difference method to analyze a buoyancy-driven flow in a cavity.

Typical layout of a data center facility

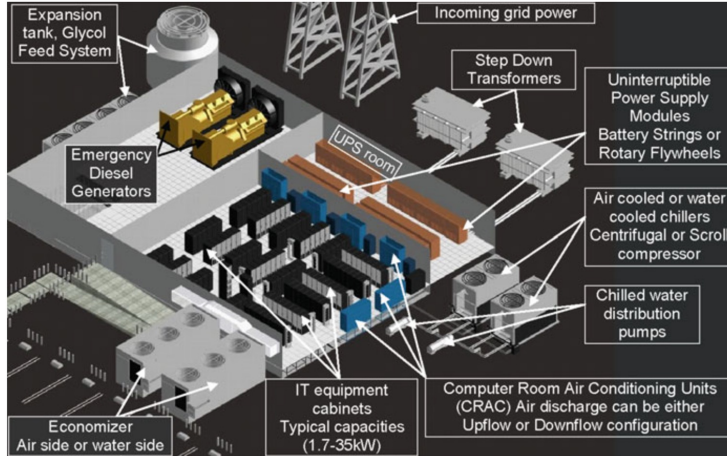


Figure 1: (Figure 1.1 in Ref. [3]).

Seminar 2

- Dr. David Geb, Synopsys (formerly Ansys)
- Title: TBD
- Date: Tuesday, 9/29/26
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Project 2

In Project 2, you will analyze airflow within the data hall using CFD. You will develop airflow handling strategies. You will use Cadence Reality DC Design Pro.

Temperature formulation of the first law of thermodynamics

This derivation comes from Ref.[\[1\]](#).

Finite Difference: Buoyance-driven Cavity

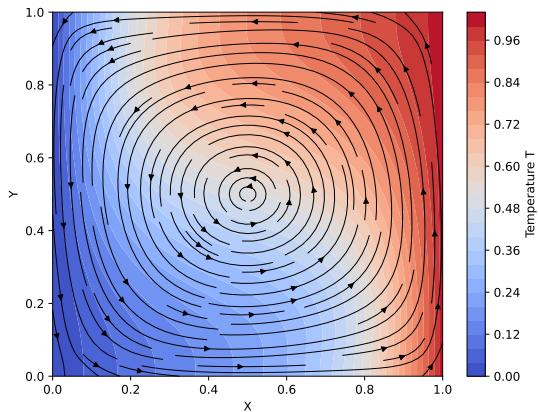


Figure 2: Compare with Figures 7.8 and 7.17 from Ref.[2].

References I

- [1] Adrian Bejan.
Convection Heat Transfer.
John Wiley & Sons, 3rd edition, 2004.
- [2] J. H. Ferziger and M. Perić.
Computational Methods for Fluid Dynamics.
Springer, 3 edition, 2002.
- [3] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.